

Margeaux Corrigan

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EDUCATION

Brown University, Sc. B. Electrical Engineering, 4.0/4.0 GPA

Providence, RI | **Expected Graduation May 2029**

Relevant Courses: Ordinary Differential Equations, Functional Programming, Statics & Dynamics, Data Structures & Algorithms

TECHNICAL SKILLS

Programming: C/C++, Python (NumPy, Pandas, Matplotlib), Ada, MATLAB, Java, OCaml, Racket

Embedded Systems: Bare-metal firmware (RP2040, STM32), register-level programming, I2C/SPI, hardware bring-up

Hardware: Schematic design, soldering, CAD & CAM (SolidWorks, Fusion 360, Onshape), FEA, 3D printing, sheet metal fabrication

RESEARCH

Interactive 3D Vision & Learning Lab (IVL), Incoming Paid Researcher

Providence, RI | Summer 2026

Advisor: Srinath Sridhar, Assistant Professor | Brown University

- Selected to investigate multi-camera capture and calibration systems using C++ and embedded hardware

INDUSTRY EXPERIENCE

AdaCore, Technical Mentorship

Providence, RI | April 2026 - Present

Mentor: Oliver Henley, UX Engineer | AdaCore

- Developed bare-metal Ada firmware for STM32G431 using direct register-level programming (RM0440/SVD), without HAL
- Parsed ARM SVD/XML files to understand register mappings, enabling faster development of STM32 firmware in Ada
- Configured STM32 GPIO peripherals directly from reference manual without HAL, validating functionality on hardware

Persistence Labs, Strategy Intern

Charleston, SC | May 2024 – October 2025

- Facilitated beta testing with high school control groups, collected feedback across 10+ iterations to identify usability issues
- Synthesized testing results into reports and presented findings to the founder/CEO, guiding UI and product design decisions
- Communicated user feedback to engineering team, influencing feature prioritization and UI design decisions

MIG3 Inc., Part-Time Technical Assistant

Irvine, CA | May 2022 – August 2025

- Provided technical support for office software and built Excel-based data tools, improving internal workflow efficiency

LEADERSHIP & ACTIVITIES

Brown Rocketry Club, Co-President, Avionics Team Lead

Providence, RI | September 2025 – Present

- Contributed to Brown Rocketry's first-ever IREC acceptance as sole avionics developer on the competing payload
- Designed IREC avionics electrical architecture and wiring diagrams for power distribution, sensor interfaces, and flight systems
- Performed FEA on avionics bay components and developed CAM-based manufacturing plans in Fusion 360
- Secured annual industry sponsorship by leading partner outreach and presenting club technical work to external stakeholders

Brown Space Engineering, Antenna Team Member

Providence, RI | March 2026 – Present

- Designing communications antennas (Nitinol dipole, copper patch), including specification and integration constraints
- Establishing feed architecture, impedance matching strategy, and EMC constraints on adjacent and connected hardware

AI & Robotics Ethics Society, Research Team Member

Providence, RI | September 2025 – Present

- Designing a survey and assessment to evaluate the effectiveness of Brown's Socially Responsible Computing curriculum
- Collaborating with the Computer Science Department and teaching assistants to collect and analyze student feedback

Project Amplify, Founder & Director

Irvine, CA | May 2024 – September 2025

- Founded a summer AI literacy program across 3+ school districts, designed and taught curriculum from scratch
- Taught students to train basic image recognition AI using Python, introducing coding and model evaluation with Pandas/NumPy

ENGINEERING PROJECTS

Open Source Bare-Metal Driver Stack, Independent Project

C | May 2026 – Present

- Developed bare-metal GPIO, SysTick, and UART drivers for RP2040 in C using direct register-level programming without HAL
- Verified UART TX via logic analyzer and serial monitor; ISR-driven SysTick timing and atomic GPIO validated
- Designed layered driver structure separating register maps from driver logic for future Cortex-M portability

Rocket Payload Sensor System, Sole Designer

RP2040, Ada | February 2026 – April 2026

- Developed bare-metal Ada firmware for an RP2040 rocket payload; work led to AdaCore mentorship
- Deployed payload on a live high-power rocket launch; logged repeatable 12+ G flight data, validating system end to end
- Integrated BMP390 and LSM9DS1 sensors over I2C with register-level configuration and data acquisition at 119 Hz
- Developed SPI flash logging pipeline for W25Q128 onboard storage; validated write/erase/read-back on hardware
- Debugged I2C, SPI, UART, and power sequencing issues across embedded hardware and firmware layers

AWARDS & GRANTS

Brown UTRA Research Award, 2026 | Brown Design Workshop Grant, 2026 | Taco Bell Ambition Accelerator Grant, 2025